A prospective multi-center quality improvement initiative (NINJA) indicates a reduction in nephrotoxic acute kidney injury in hospitalized children

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Nephrotoxic medication (NTMx) exposure is a common cause of acute kidney injury (AKI) in hospitalized children. The Nephrotoxic Injury Negated by Just-in time Action (NINJA) program decreased NTMx associated AKI (NTMx-AKI) by 62% at one center. To further test the program, we incorporated NINJA across nine centers with the goal of reducing NTMx exposure and, consequently, AKI rates across these centers. NINJA screens all non-critically ill hospitalized patients for high NTMx exposure (over three medications on the same day or an intravenous aminoglycoside over three consecutive days), and then recommends obtaining a daily serum creatinine level in exposed patients for the duration of, and two days after, exposure ending. Additionally, substitution of equally efficacious but less nephrotoxic medications for exposed patients starting the day of exposure was recommended when possible. The main outcome was AKI as defined by the Kidney Disease Improving Global Outcomes (KDIGO) serum creatinine criteria (increase of 50% or 0.3 mg/dl over baseline). The primary outcome measure was AKI episodes per 1000 patient-days. Improvement was defined by statistical process control methodology and confirmed by Autoregressive Integrated Moving Average (ARIMA) modeling. Eight consecutive bi-weekly measure rates in the same direction from the established baseline qualified as special cause change for special process control. We observed a significant and sustained 23.8% decrease in NTMx-AKI rates by statistical process control analysis and by ARIMA modeling; similar to those of the pilot single center. Thus, we have successfully applied the NINJA program to multiple pediatric institutions yielding decreased AKI rates.

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Nephrotoxic medication exposure is one of the most common causes of acute kidney injury (AKI) in hospitalized children.[1] More than 80% of non–critically ill hospitalized children receive a nephrotoxic medication,[2] and the associated AKI leads to increased hospital length of stay and cost ($14,000–$17,000 per event).[3,4] Furthermore, 10%-49% of adults and children who survive AKI develop chronic kidney disease,[5–8] including those with isolated nephrotoxic medication–associated AKI.[9]

Nephrotoxic medication–associated AKI affects 19%–31% of children who receive an intravenous aminoglycoside.[3] The AKI rate doubles when children receive 3 or more nephrotoxic medications on the same day.[2] Although monitoring kidney function with serum creatinine assessment is standard practice to detect kidney function decline, serum creatinine was assessed at least every 4 days only 50%–60% of the time in these studies. It is not surprising then, that nephrotoxic medication–associated AKI goes unrecognized; 63% of children with nephrotoxic medication–associated AKI, as determined by increases in serum creatinine, do not have AKI listed as a diagnosis in their electronic health record.[10]

Based on these observations, a single pediatric center instituted a systematic screening program, called Nephrotoxic Injury Negated by Just-in-Time Action (NINJA), to assess for AKI development in non–critically ill hospitalized children exposed to a high nephrotoxic medication burden.[11] This center realized sustained reductions in nephrotoxic medication exposure and AKI rates for the 3.5 years after the program's inception.[12]

Unfortunately, effective single-center harm-reduction strategies do not universally succeed at other institutions, and this lack of success may be related to a number of factors, including the reliability of implementation of the strategies and the initial baseline prevalence or incidence of the condition. For the latter, it is likely that centers with a relatively high prevalence of the condition are more likely to see incremental gains with an intervention than centers with relatively low rates. Since the release of the Institute of Medicine report, *To Err Is Human*, substantial hospital, state, and federal resources have been directed toward the study and implementation of patient safety initiatives.[13] Although individual hospitals have observed improvements in patient safety, it has been uncommon to see success spread beyond discrete early adopter hospitals.[14–17] We established a collaborative of 9 pediatric institutions to disseminate the program and test the hypothesis that our approach would decrease nephrotoxic medication–associated AKI beyond the single-center experience.

RESULTS

Collaborative data

The collaborative included 638,695 inpatient hospital days, from July 2015 through June 2017. During these days, 4513 exposed patients experienced 746 AKI episodes. We observed improvement in AKI prevalence rates from 1.7 to 1.3 episodes per 1000 patient-days (Figure 1; 23.8% reduction) and AKI rates per exposure from 23.6% to 15.0% (Figure 2; 36.7% reduction). The AKI reduction was confirmed by autoregressive integrated moving average (ARIMA) modeling—there was a significant decline in AKI rates, from 1.40 to 0.96 per 1000 patient-days, resulting in an average decline of 8 patients with AKI per 1000 patient-days throughout the study period (95% confidence interval: –1.44 to –14.57; $P = 0.018$; Figure 3).

Both AKI rate reductions were sustained, despite variation in exposure rates (Figure 4) and sustained increases in the collaborative composite maturity score. The exposure rate initially decreased, then increased with increasing composite maturity, and then decreased again from that new elevated rate. We did not observe an improvement in the AKI intensity rate (Figure 5); however, the collaborative baseline rate was

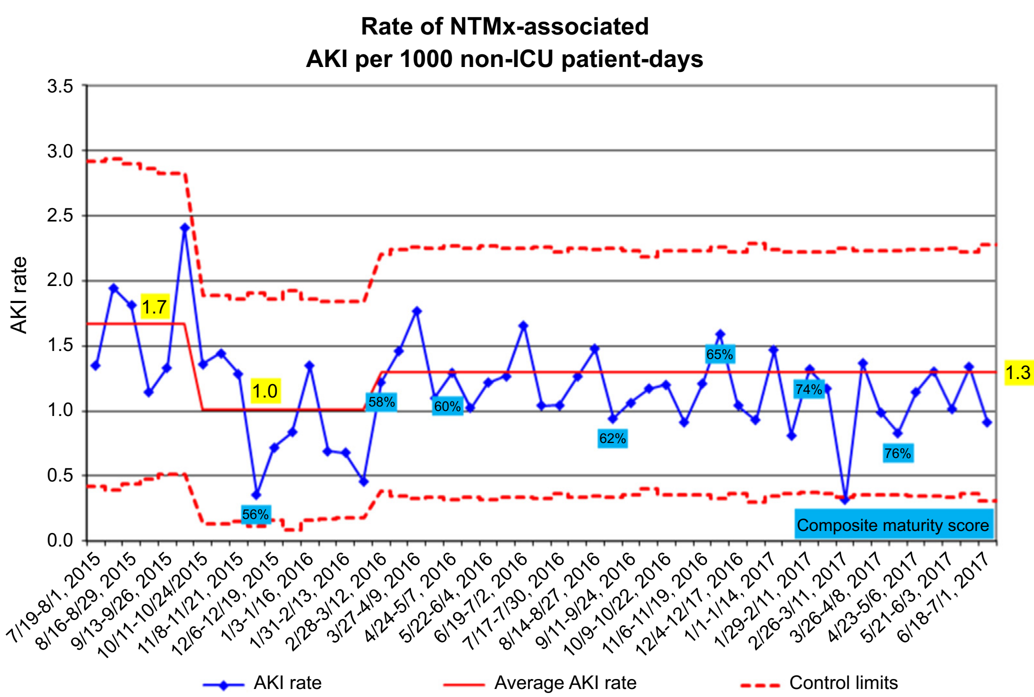

Figure 1 | Biweekly average acute kidney injury (AKI) development rates as measured by patient number with AKI per 1000 non–critically ill patient hospital days. Although some centers started submitting data in October 2014, all 9 centers were submitting data reliably in July 2015, which is when the baseline rate was established for purposes of this analysis. The AKI rates decreased from 1.7 to 1.3 AKI per 1000 patient-days over the course of the study as revealed by 8 consecutive weekly rates below the baseline rate, representing a 99.7% likelihood that a special cause was present. Each data point represents a 14-day duration. The percentages noted in the blue boxes reflect the composite maturity score for the collaborative as calculated in Supplementary Table S4. ICU, intensive care unit; NTMx, nephrotoxic medication.

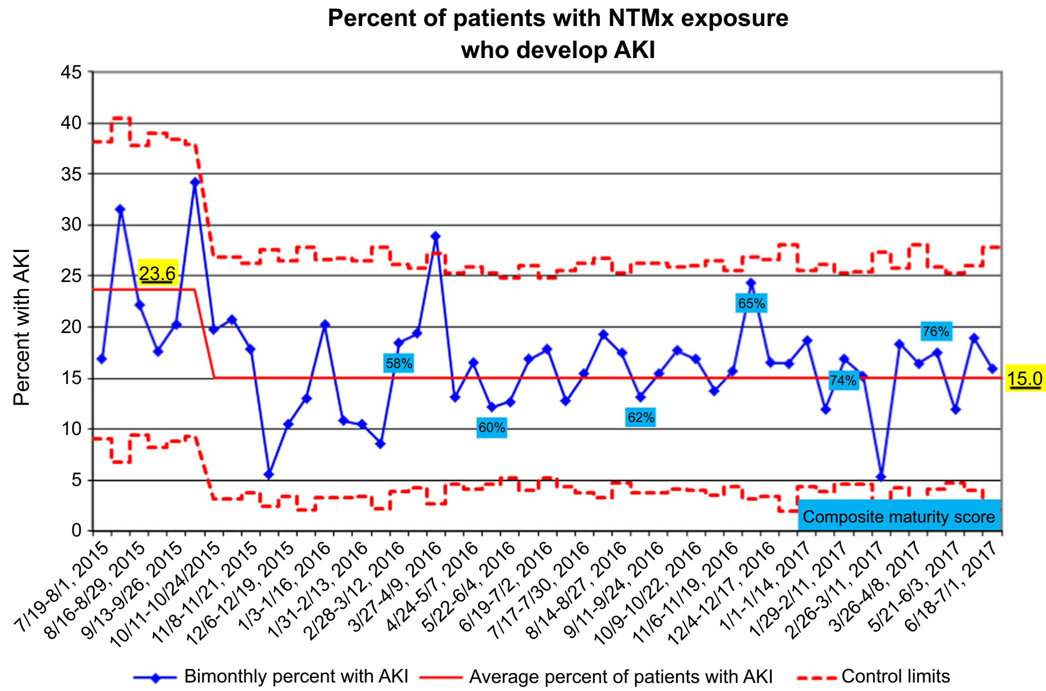

Figure 2 | Biweekly average acute kidney injury (AKI) development rates as measured as a percentage of the patient population exposed to nephrotoxic medication (NTMx). Although some centers started submitting data in October 2014, all 9 centers were submitting data reliably in July 2015, which is when the baseline rate was established for purposes of this analysis. These AKI rates decreased from 23.6% to 15.0% over the course of the study, as revealed by 8 consecutive weekly rates below the baseline rate, representing a 99.7% likelihood that a special cause was present. Each data point represents a 14-day duration. The percentages noted in the blue boxes reflect the composite implementation score for the collaborative as calculated in Supplementary Table S4.

almost half that of the single-center study. The rates achieved in the other 3 measures were similar to the final rates previously reported from the single center (Table 1).[12] Assuming the initial baseline AKI rates would have persisted without program implementation, we estimate that 242 episodes of AKI were avoided (Table 2) over the 2-year course of study.

Collaborative and individual-center assessments

The baseline and final exposure and AKI rates for the collaborative and each center are depicted in Supplementary Figure S1A and B. The statistical process control charts for exposure and AKI rates are depicted for each center in Supplementary Figure S2A and B. Three centers had lower

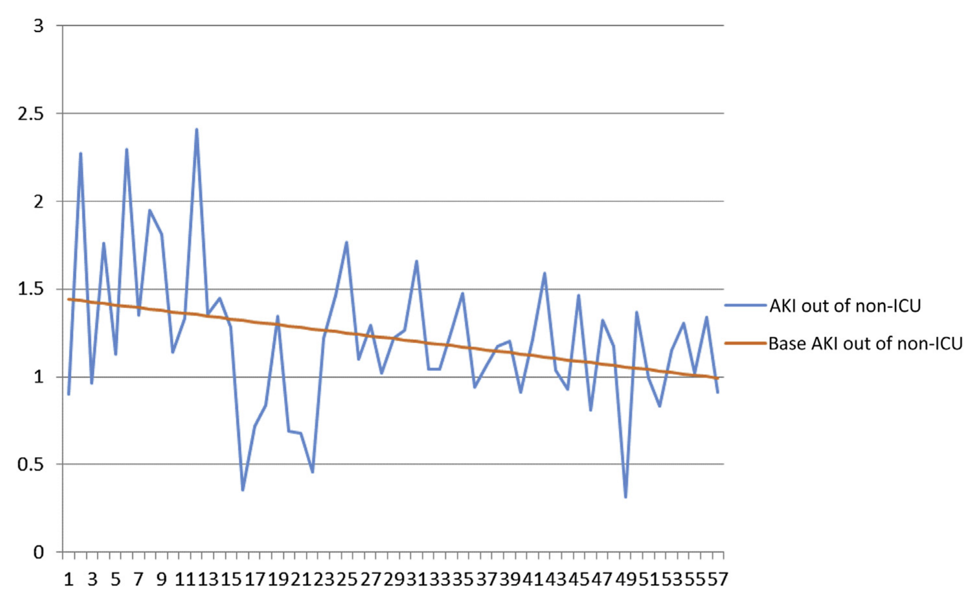

Figure 3 | Autoregressive integrated moving average (ARIMA) modeling methods are given in the Supplementary Methods. AKI, acute kidney injury; ICU, intensive care unit.

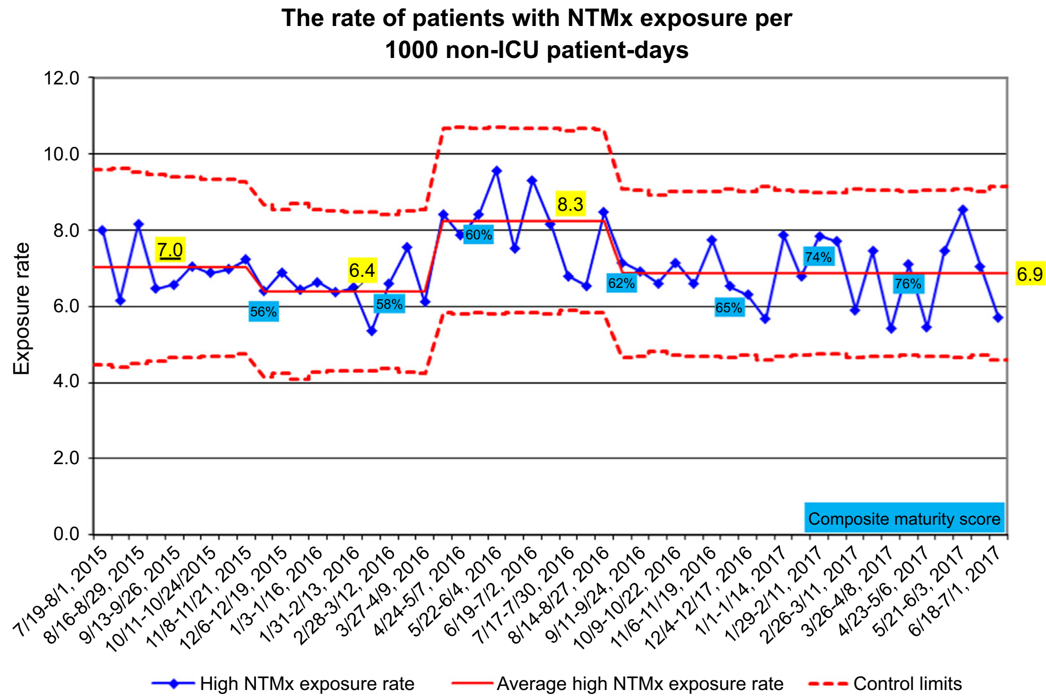

Figure 4 | Biweekly average nephrotoxic medication exposure rates as measured by exposed patients per 1000 non–critically ill patient hospital days. Although some centers started submitting data in October 2014, all 9 centers were submitting data reliably in July 2015, which is when the baseline rate was established for purposes of this analysis. This exposure rate changed multiple times over the course of the study, as revealed by 8 consecutive weekly rates above or below each established rate, representing a 99.7% likelihood that a special cause was present. Each data point represents a 14-day duration. The percentages noted in the blue boxes reflect the composite implementation score for the collaborative as calculated in Supplementary Table S4. ICU, intensive care unit; NTMx, nephrotoxic medication.

baseline AKI rates than the collaborative post-implementation study rate of 1.3 episodes per 1000 patient-days; none of these centers demonstrated improvement after implementation. Five of 6 centers with baseline AKI rates above this collaborative post-implementation rate demonstrated improvement, with 3 achieving rates below the final collaborative AKI rate.

The collaborative demonstrated increasing implementation maturity in each process measure (Figure 6). Each center's progress in implementation is depicted in Supplementary Table S1. Although all of the centers were submitting data reliably by July 2015, Supplementary Table S1 shows that the maturity of, or compliance with, the execution of each component occurred at different times for each center. Only one center (C7) had a program in place to detect nephrotoxic medication exposure and AKI prior to its collaborative participation. However, this center did not use the same exposure rules prior to the current work.

DISCUSSION

This multi-center initiative focusing on nephrotoxic medication exposure in non–critically ill hospitalized children led to sustained reductions in nephrotoxic medication–associated AKI rates. The post-implementation collaborative rates achieved after dissemination were the same or lower than those achieved in the single center that pioneered the program. All but 3 centers maintained or achieved an AKI rate below 1.3 episodes per 1000 patient-days, and 5 of 6 with baseline rates above this threshold reduced their rates over the course of the study.

Our study demonstrates that some nephrotoxic medication burden may be avoidable, and medication-related AKI should be considered a potentially modifiable adverse event. Although nephrotoxic medications are often needed in hospitalized patients, the program's trigger to alert the healthcare team to the increasing nephrotoxic medication burden likely focused attention on adjudicating which medications were necessary and which could be electively discontinued or substituted for with a less nephrotoxic alternative.

This collaborative work revealed several novel findings. As opposed to the single-center study, where the program was implemented reliably in all service lines, and with serum creatinine assessed daily in >99% of exposed patients, the collaborative started with a lower level of participation and less-consistent serum creatinine assessment. Exposure rates increased as more service lines began participation, but then they decreased again with stable program implementation. Decreases in AKI rates were sustained, even with improved implementation. The sustained AKI rate reductions observed suggest that centers not only increased detection of potentially harmful exposures, but also maintained vigilance on nephrotoxic medication burden and associated AKI. We do note that although all centers increased serum creatinine assessment, only one center achieved 100% adherence to detection

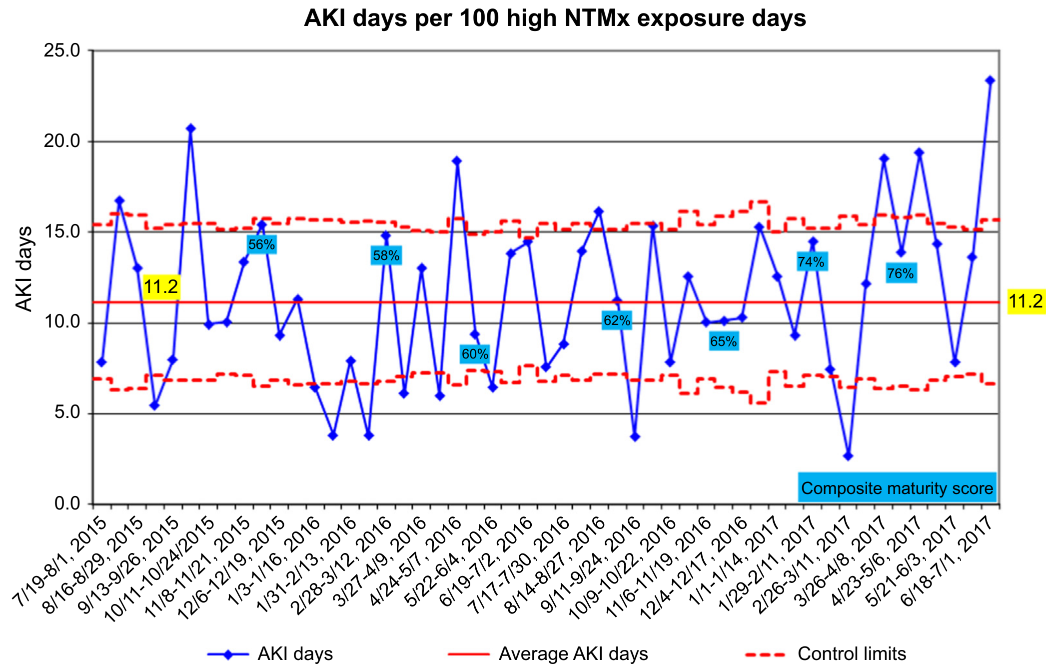

Figure 5 | Biweekly average acute kidney injury (AKI) development rates as measured as a percentage of the patient population exposed to nephrotoxic medication (NTMx). Although some centers started submitting data in October 2014, all 9 centers were submitting data reliably in July 2015, which is when the baseline rate was established for purposes of this analysis. This exposure rate remained constant throughout the course of the study. The percentages noted in the blue boxes reflect the composite implementation score for the collaborative as calculated in Supplementary Table S4.

in exposed patients. The strategy employed by centers to improve serum creatinine assessment included presentation of data from their own center and comparative collaborative data on exposure, AKI, and serum creatinine ascertainment to key stakeholders at the administrative and service line levels. Although all centers improved, it is clear that strategies other than a shared collaborative data transparency are needed to gain 100% adherence to this practice standard.

Another novel outcome is the observed individual center AKI rate changes from the beginning to the end of the observation period. The fact that 5 of 6 centers that began with an AKI rate higher than 1.3 episodes per 1000 patient-days achieved a lower rate, and that 3 of these achieved a rate at or near this post-implementation rate, suggests that this threshold may be a reliable benchmark for future institutions. The differences in the time to achieve and sustain reductions in AKI rates suggest that there may be unique contextual factors at some institutions that enabled rapid implementation. Understanding these factors may help other hospitals optimize their context prior to implementation. The observation that all 3 centers that started with AKI rates below the end-of-study thresholds saw no improvement

Table 1 | Mean (95% CI) baseline and end measure rates[a] for the collaborative and the comparative pilot single center

Measure	Collaborative baseline (dates)	Collaborative end (dates)	Collaborative change (%)	Single pilot center comparator[b]
AKI rate[c] (per 1000 patient-days)	1.7 (0.8–3.0) (July 19, 2015–October 10, 2015)	1.3 (0.6–1.8) (February 14, 2016–July 1, 2017)	−23.5 (−54.4 to 23.5)	1.1
Exposure rate[d] (per 1000 patient-days)	7.0 (6.1–9.4) (July 19, 2015–November 22, 2015)	6.9 (5.1–11.1) (September 11, 2016–July 1, 2017)	−1.4 (−24.4 to 30.4)	7.5
AKI rate (per exposure)	23.8% (13.6%–30.4%) (July 19, 2015–October 10, 2015)	15.0% (8.5%–24.7%) (October 11, 2015–July 1, 2017)	−36.4 (−59.4 to 29.7)	15.4%
AKI intensity rate (AKI days/100 exposure days)	11.3 (7.5–22.7) (July 19, 2015–July 1, 2017)	11.3 (6.5–13.5) (July 19, 2015–July 1, 2017)	0 (−64.6 to 54.2)	19.1

AKI, acute kidney injury; CI, confidence interval.
[a]Data span from July 2015—when all 9 centers were transmitting data reliably—through June 2017.
[b]Based on the rates seen at the end of previously published study.[12]
[c]The acute kidney injury rate per 1000 patient-days changed twice during the course of the study; only the initial and end rates are shown. See Figure 1 for detailed depiction of biweekly rates.
[d]The exposure rate changed 3 times during the course of the study; only the initial and end rates are shown. See Supplementary Figure S2 for the detailed depiction of biweekly rates.

Table 2 | AKI episodes avoided

Measure	Baseline initial era (July 2015–September 2015)	Intermediate NINJA era (October 2015–February 2016)	Mature NINJA implementation era (March 2016–June 2017)
Non–critically ill patient-days	61,798	122,458	454,439
Number of patients expected to have AKI with initial AKI rate	NA	208	773
Number of patients with AKI	103	112	531
Patients with AKI avoided	NA	96	242

AKI, acute kidney injury; NA, not applicable; NINJA, Nephrotoxic Injury Negated by Just-in-Time Action.

indicates that these thresholds may be the limit of what can be achieved solely by sharing data, following best practices, and gaining process reliability. This is not to say that centers that begin with AKI rates below this benchmark should not use the program, as centers can't know their rates without using such a program. Center C5 represents an unusual example in that it started and ended with an AKI rate of 1.6 per 1000 patient-days, above the post-implementation rate. What is interesting about this center is that it demonstrated the highest maturity in all components of the NINJA process, except for improvement in daily serum creatinine measurement, which stayed at a level of 2 out of 4. So, it is possible that the AKI rates reported over the course of the study are subject to an ascertainment bias that would lead to an inability to detect a change. Center C8 is an informative example, where initial daily serum creatinine assessment and service line implementation were low. As this center increased internal dissemination, exposure, and AKI detection, the corresponding rates increased initially, but improved soon thereafter. The decrease in variation in outcome rates across centers adds credibility to the possibility that process reliability is critical to avoidance of preventable harm. We suggest that achieving *further* harm reduction requires innovation beyond standardization of current best practice.

As noted in Table 1, exposure rates did not change from study beginning to the post-implementation period. It may seem surprising, then, that the AKI rate among the exposed patients decreased. We suggest that this finding may show strong benefit of the intervention. Although exposure and AKI rates per 1000 patient-days are intertwined, the AKI rates per exposure and intensity rates are not. We speculate that different medication combinations or an earlier reduction in nephrotoxic medication exposure may have led to these outcomes, but we do not have the data to support this conclusion.

Our study provides a contribution to the published literature of quality improvement projects that leverage information in the electronic health record to improve outcomes in adult patients with, or at-risk for, AKI. Early studies reported on programs that would alert physicians to adjust medications or assess fluid status in patients with

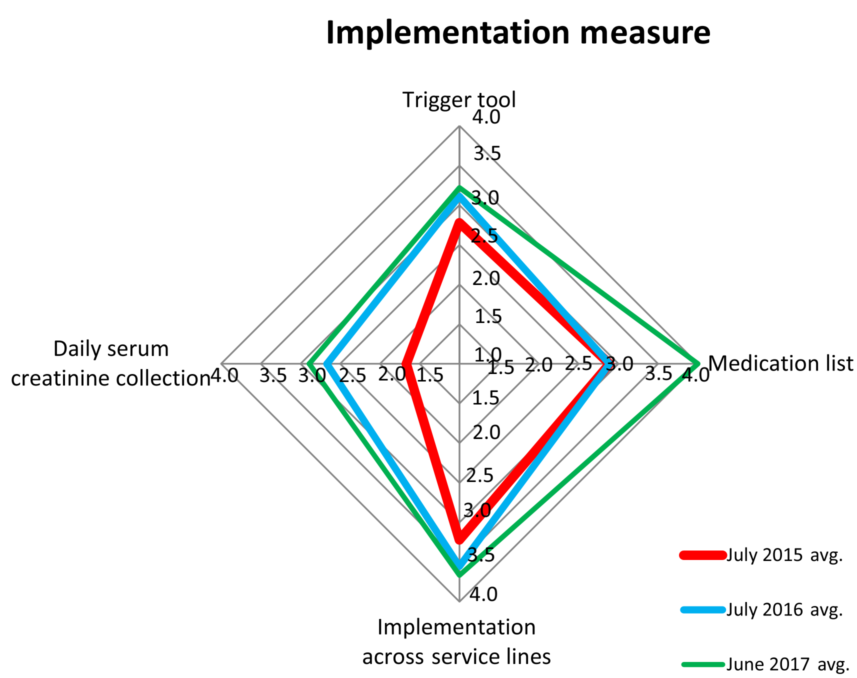

Figure 6 | These components of the composite implementation score for the collaborative are calculated in Supplementary Table S4. avg., average.

AKI.[18,19] In these studies, physician response and intervention time decreased after the AKI alert was put in place. As we discussed previously,[12] our approach and form of clinical decision support have been successful, whereas a recent attempt to provide AKI-related clinical decision support to care providers has shown no significant clinical effects and may have, in fact, caused increased healthcare utilization and expenditures.[20] There are many differences that may have led to these different outcomes. First and foremost, our approach of risk-stratifying patients and providing clinical decision support upstream of AKI development provided a larger window of opportunity for elimination or mitigation through earlier clinical action. Second, we achieved our improvement in outcomes without prescribing or mandating any intervention other than asking providers to order surveillance serum creatinine measurements. The clinical pharmacists embedded in the care teams most certainly did provide guidance, but this was not standardized. Finally, a recent quality improvement program was implemented sequentially in 5 centers in the United Kingdom to determine if a complex intervention for AKI (comprising AKI e-alerts, an AKI care bundle, and a program of AKI education) would improve standards of care delivery and lead to better patient outcomes.[21] One of the core approaches of the AKI bundle was to "review medications and stop those contributing to AKI." These investigators observed increased recognition of AKI and decreased hospital length of stay after implementation of their program.

Despite the strengths of a multi-center collaborative and the confirmatory nature of the outcomes achieved, this study has a number of limitations. First, we cannot extrapolate our data to critically ill patients, as AKI is usually multifactorial in the intensive care unit. Second, it is possible that some patients had AKI resulting from causes other than nephrotoxic medication exposure. We suggest that this possibility does not invalidate the benefit of the program. We detected a high rate of AKI in exposed patients, which would lead to appropriate interventions irrespective of cause, a strategy recommended by the Kidney Disease: Improving Global Outcomes (KDIGO) AKI Guidelines.[22] Third, although we can only speculate as to the reasons for changes in our observed measures, other factors—including specific combinations of medications, rates of underlying chronic kidney disease, volume depletion, and genetic predisposition to nephrotoxic medication–associated AKI—could confound any attribution to improvement, but these were either not part of the intervention, were not systematically assessed, or are not modifiable. Fifth, we acknowledge differences in the serum creatinine surveillance rates across sites and across service lines. Sixth, we acknowledge a limitation of widespread generalizability of our approach. For example, each of the 9 centers in the study had a pediatric nephrologist with prior research experience in AKI, which most inpatient pediatric facilities do not have. Finally, it is possible that we had some unanticipated consequences of decreasing the nephrotoxic medication burden. For example, we could have caused decreased infection eradication rates if the medications chosen over nephrotoxic medications were less efficacious, although such a phenomenon was not observed in the single-center study.[12]

We demonstrated successful dissemination-to-implementation steps of a program to decrease nephrotoxic medication–associated AKI in children at 9 pediatric institutions. Given the ubiquitous administration of nephrotoxic medications to hospitalized individuals, we speculate that more widespread dissemination to other pediatric and adult centers could lead to decreased patient morbidity and reduction in associated healthcare costs related to AKI. Gaining a deeper understanding of the conditions that supported implementation at these 9 sites is an important next step in developing a robust approach to spread the program more broadly across a diverse range of hospitals.

METHODS

Institution participants

The institutions were chosen to represent a diversity of structure (independent, freestanding pediatric hospital vs. a "children's hospital within a hospital"), bed capacity, and geographic location (Supplementary Table S2). All institutions had a pediatric nephrologist champion who had previously participated in a multi-center pediatric AKI related study. The project was approved by all institutional review boards, with a waiver of patient/parental informed consent as the daily serum creatinine assessment intervention was considered standard of care in patients with high nephrotoxic medication exposure.

The intervention

The intervention is described in detail elsewhere.[11] Each center established a weekday screening method to identify non–critically ill patients with high nephrotoxic medication exposure ("exposure"; defined below). Patients admitted to intensive care units were excluded, as AKI is multifactorial in critically ill children.[23–26] Patients with urinary tract infections were excluded, as AKI can be associated with infection itself. For the intervention, site team members recommended daily serum creatinine monitoring or substitution of equally efficacious but less-nephrotoxic medication for exposed patients starting on the day of exposure. Screening did not occur on weekends at all sites.

Dissemination model

In 2014, we hosted a 2-day learning session to outline the strategies and measures used to implement the program, which was grounded in the Institute for Healthcare Improvement Breakthrough Series approach.[27] Each institution committed to implementing a number of interventions (Supplementary Table S3), including daily serum creatinine monitoring in patients exposed to a high nephrotoxic medication burden. We used monthly web-based teleconferences and virtual learning supplemented by annual face-to-face meetings to support sites in the system and process changes necessary to implement the program.

Implementation measure

Reliable, hospital system–wide implementation of the project requires: (i) a "trigger tool" to detect high nephrotoxic medication exposure in near real-time (usually embedded in the electronic health record); (ii) adoption of the complete collaborative nephrotoxic medication list; (iii) daily serum creatinine assessment in

Table 3 | Program outcome measures

Measure name	Numerator	Denominator	Clinical meaning
AKI prevalence rate (per 1000 patient-days)	Number of patients with high nephrotoxic medication exposure[a] who developed AKI in the calendar week of study	The total number of non–critically ill patient hospital days standardized per 1000 patient-days in the calendar week of study	This measure generates a normalized rate of AKI cases per study week. *Decreases in this measure could result from decreases in AKI or decreases in high nephrotoxic medication exposure.*
High nephrotoxic medication exposure[a] prevalence rate (per 1000 patient-days)	Number of new patients with high nephrotoxic medication exposure in the calendar week of study	The total number of non–critically ill patient hospital days standardized per 1000 patient-days in the calendar week of study	This measure generates a normalized rate of high nephrotoxic medication exposure cases per study week. *Decreases in this measure result from decreases in high nephrotoxic medication burden in the non–critically ill population.*
Rate of patients with high nephrotoxic medication exposure who develop AKI (%)	Number of patients who develop AKI[b]	Number of new patients with high nephrotoxic medication exposure in the calendar week of study	This measure generates the fraction of patients with high nephrotoxic medication exposure who develop AKI. *This measure normalizes the AKI rate by the high nephrotoxic medication exposure rate.*
AKI intensity rate (per 100 exposed patient-days)	Number of days patients have AKI	The total number of exposed patient-days standardized per 100 exposed days	This measure depicts a normalized duration of AKI per exposed days. *This rate measures how sustained the AKI episodes are.*

AKI, acute kidney injury.
[a]High nephrotoxic medication exposure is defined as receipt of ≥3 nephrotoxic medications on the same calendar day, or ≥3 consecutive days of an intravenous aminoglycoside.
[b]AKI development factors into the numerator of the biweekly time period during which the patient became exposed, even if AKI develops in a different calendar week than when a patient became exposed.

exposed patients; and (iv) data capture from all general, subspecialty, and surgical service lines. We developed a composite score to monitor the implementation of these key program components and examined each of the components separately (Supplementary Table S4). The composite measure was tracked quarterly at each center and across the collaborative.

Operational definitions and outcome measures

High nephrotoxic medication exposure ("exposure"). Exposure was defined as receipt of an intravenous aminoglycoside on ≥3 consecutive days or ≥3 nephrotoxic medications on the same calendar day. Patients were considered exposed during the time of exposure and for the 2 days after exposure ended. The list of nephrotoxic medications was derived from that used in the previous study (Supplementary Table S5)[11] and was updated in March 2016 by a committee from the collaborative, based on new evidence from the literature (Supplementary Table S6).

AKI. AKI was defined by the Kidney Disease: Improving Global Outcomes (KDIGO) consensus criteria—as an increase in serum creatinine of either 50% occurring within 7 days *or* 0.3 mg/dl occurring within 48 hours, compared to the lowest value obtained within the previous 6 months, which served as a measured baeseline.[22] If a baseline serum creatinine level was unavailable from within the previous 6 months, it was imputed based on a presumed estimated creatinine clearance of 120 ml/min per 1.73 m^2, and solving for serum creatinine using the formula

$$\text{serum creatinine (mg/dl)} = 0.413 \times \text{patient height (cm)}/120,$$

as validated in the pediatric literature.[28,29] In order to improve the specificity of AKI diagnosis, we required that the serum creatinine increase should reach a concentration of at least 0.5 mg/dl. Otherwise, small, likely clinically irrelevant increases (e.g., from 0.2 to 0.3 mg/dl) would meet the diagnosis. The AKI duration was calculated in days from the day of onset to resolution, defined as when the serum creatinine level returned to baseline for 2 consecutive days or after 30 days of persistent AKI, whichever came first. The KDIGO urine output criteria were not used, because nephrotoxic AKI is usually non-oliguric.[30]

Outcome measures. The collaborative adopted the 4 outcome measures used in the single-center report (Table 3), with the primary outcome measure being the number of exposed patients with AKI per 1000 non–critically ill patient hospital days.[11,12] Each center submitted de-identified data aggregated on a biweekly basis to a password-protected central website.

Statistical analysis

Sites began submitting data in October 2014. In July 2015, all 9 centers were transmitting data reliably, and this was considered the time point to set baseline rates for the outcome measures. We report data from July 2015 through June 2017 for the present analysis.

We used the same statistical process control methodology[31] employed in the single-center study to identify changes from baseline for each measure. We set an *a priori* standard of 8 consecutive biweekly measure rates in the same direction, either above or below the established baseline rate, to qualify as a statistical change, which corresponds to a 99.7% likelihood that the change resulted from the intervention.[32] This methodology served as the primary quality improvement assessment measurement to track serious safety event rates for the Solutions for Patient Safety Network, a national pediatric harm-reduction collaborative.[33,34] To validate statistical process control results, we also performed autoregressive integrated moving average modeling; description of the modeling methodology is given in the Supplementary Methods.

DISCLOSURE

SLG and ESK receive royalties from Vigilanz Corporation (Minneapolis, MN) for the Nephrotoxic Injury Negated by Just-in-Time (NINJA) application licensed to Vigilanz from Cincinnati Children's Hospital Medical Center. The Vigilanz application was not developed or used for the current study. All the other authors declared no competing interests.

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SUPPLEMENTARY MATERIAL

Supplementary File (PDF)

Figure S1A. Collaborative baseline and post-intervention exposure and acute kidney injury rates.

Figure S1B. Individual center baseline and post-intervention exposure and acute kidney injury rates.

Figure S2A. Individual center statistical process control charts for acute kidney injury per 1000 non-ICU patient-days.

Figure S2B. Individual center statistical process control charts for nephrotoxic medication exposure per 1000 non-ICU patient-days.

Table S1. Individual center progress of the NINJA implementation scoring system.

Table S2. Hospital characteristics in the NINJA collaborative.

Table S3. Commitments to participate in the NINJA collaborative.

Table S4. NINJA implementation scoring system.

Table S5. Initial list of nephrotoxic medications (45 med list).

Table S6. Amended list of nephrotoxic medications (61 med list).

Supplementary Methods. ARIMA modeling.

REFERENCES

1. Hui-Stickle S, Brewer ED, Goldstein SL. Pediatric ARF epidemiology at a tertiary care center from 1999 to 2001. *Am J Kidney Dis*. 2005;45:96–101.
2. Moffett BS, Goldstein SL. Acute kidney injury and increasing nephrotoxic-medication exposure in noncritically-ill children. *Clin J Am Soc Nephrol*. 2011;6:856–863.
3. Zappitelli M, Moffett BS, Hyder A, et al. Acute kidney injury in non-critically ill children treated with aminoglycoside antibiotics in a tertiary healthcare centre: a retrospective cohort study. *Nephrol Dial Transplant*. 2011;26:144–150.
4. Misurac JM, Knoderer CA, Leiser JD, et al. Nonsteroidal anti-inflammatory drugs are an important cause of acute kidney injury in children. *J Pediatr*. 2013;162:1153–1159, 1159 e1151.
5. Askenazi DJ, Feig DI, Graham NM, et al. 3-5 year longitudinal follow-up of pediatric patients after acute renal failure. *Kidney Int*. 2006;69:184–189.
6. Mammen C, Al Abbas A, Skippen P, et al. Long-term risk of CKD in children surviving episodes of acute kidney injury in the intensive care unit: a prospective cohort study. *Am J Kidney Dis*. 2012;59:523–530.
7. Goldstein SL, Jaber BL, Faubel S, et al. AKI transition of care: a potential opportunity to detect and prevent CKD. *Clin J Am Soc Nephrol*. 2013;8:476–483.
8. Garg AX, Suri RS, Barrowman N, et al. Long-term renal prognosis of diarrhea-associated hemolytic uremic syndrome: a systematic review, meta-analysis, and meta-regression. *JAMA*. 2003;290:1360–1370.
9. Menon S, Kirkendall ES, Nguyen H, et al. Acute kidney injury associated with high nephrotoxic medication exposure leads to chronic kidney disease after 6 months. *J Pediatr*. 2014;165:522–527 e522.
10. Schaffzin JK, Dodd CN, Nguyen H, et al. Administrative data misclassifies and fails to identify nephrotoxin associated acute kidney injury in hospitalized children. *Hosp Pediatr*. 2014;4:159–166.
11. Goldstein SL, Kirkendall E, Nguyen H, et al. Electronic health record identification of nephrotoxin exposure and associated acute kidney injury. *Pediatrics*. 2013;132:e756–e767.
12. Goldstein SL, Mottes T, Simpson K, et al. A sustained quality improvement program reduces nephrotoxic medication-associated acute kidney injury. *Kidney Int*. 2016;90:212–221.
13. Institute of Medicine (US) Committee on Quality of Health Care in America; Kohn LT, Corrigan JM, Donaldson MS, eds. *To Err is Human: Building a Safer Healthcare System*. Washington, DC: National Academies Press; 2000.
14. Walsh KE, Bundy DG, Landrigan CP. Preventing health care-associated harm in children. *JAMA*. 2014;311:1731–1732.
15. Leape LL, Berwick DM. Five years after *To Err Is Human*: What have we learned? *JAMA*. 2005;293:2384–2390.
16. Landrigan CP, Parry GJ, Bones CB, et al. Temporal trends in rates of patient harm resulting from medical care. *N Engl J Med*. 2010;363:2124–2134.
17. Shekelle PG, Pronovost PJ, Wachter RM, et al. *Assessing the Evidence for Context-Sensitive Effectiveness and Safety of Patient Safety Practices: Developing Criteria*. Rockville, MD: Agency for Healthcare Research and Quality; 2010.
18. McCoy AB, Waitman LR, Gadd CS, et al. A computerized provider order entry intervention for medication safety during acute kidney injury: a quality improvement report. *Am J Kidney Dis*. 2010;56:832–841.
19. Colpaert K, Hoste EA, Steurbaut K, et al. Impact of real-time electronic alerting of acute kidney injury on therapeutic intervention and progression of RIFLE class. *Crit Care Med*. 2012;40:1164–1170.
20. Wilson FP, Shashaty M, Testani J, et al. Automated, electronic alerts for acute kidney injury: a single-blind, parallel-group, randomised controlled trial. *Lancet*. 2015;385:1966–1974.
21. Selby NM, Casula A, Lamming L, et al. An organizational-level program of intervention for AKI: a pragmatic stepped wedge cluster randomized trial. *J Am Soc Nephrol*. 2019;30:505–515.
22. Kidney Disease: Improving Global Outcomes (KDIGO) Acute Kidney Injury Work Group. KDIGO clinical practice guideline for acute kidney injury. *Kidney Int Suppl*. 2012;(2):1–138.
23. Vachvanichsanong P, Dissaneewate P, Lim A, et al. Childhood acute renal failure: 22-year experience in a university hospital in southern Thailand. *Pediatrics*. 2006;118:e786–e791.
24. Akcan-Arikan A, Zappitelli M, Loftis LL, et al. Modified RIFLE criteria in critically ill children with acute kidney injury. *Kidney Int*. 2007;71:1028–1035.
25. Symons JM, Chua AN, Somers MJ, et al. Demographic characteristics of pediatric continuous renal replacement therapy: a report of the prospective pediatric continuous renal replacement therapy registry. *Clin J Am Soc Nephrol*. 2007;2:732–738.
26. Schneider J, Khemani R, Grushkin C, et al. Serum creatinine as stratified in the RIFLE score for acute kidney injury is associated with mortality and length of stay for children in the pediatric intensive care unit. *Crit Care Med*. 2010;38:933–939.
27. Kilo CM. A framework for collaborative improvement: lessons from the Institute for Healthcare Improvement's Breakthrough Series. *Qual Manag Health Care*. 1998;6:1–13.
28. Zappitelli M, Zhang X, Foster BJ. Estimating glomerular filtration rate in children at serial follow-up when height is unknown. *Clin J Am Soc Nephrol*. 2010;5:1763–1769.
29. Kaddourah A, Basu RK, Bagshaw SM, et al. Epidemiology of acute kidney injury in critically ill children and young adults. *N Engl J Med*. 2017;376:11–20.
30. Schetz M, Dasta J, Goldstein S, et al. Drug-induced acute kidney injury. *Curr Opin Crit Care*. 2005;11:555–565.
31. Langley GJ, Moen RD, Nolan KM, et al. *The Improvement Guide: A Practical Approach to Enhancing Organizational Performance*. 2nd ed. San Francisco, CA: Jossey-Bass; 2009.
32. Mohammed MA, Worthington P, Woodall WH. Plotting basic control charts: tutorial notes for healthcare practitioners. *Qual Saf Health Care*. 2008;17:137–145.
33. Muething SE, Goudie A, Schoettker PJ, et al. Quality improvement initiative to reduce serious safety events and improve patient safety culture. *Pediatrics*. 2012;130:e423–e431.
34. Schaffzin JK, Harte L, Marquette S, et al. Surgical site infection reduction by the Solutions for Patient Safety Hospital Engagement Network. *Pediatrics*. 2015;136:e1353–e1360.